EmoShapelets: Capturing Local Dynamics of Audio-visual Affective Speech

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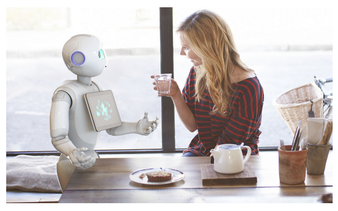
Introduction

Accompanying the growing prevalence of human-computer interactions is the increasing need for computer agents and health-care automatons to understand human emotions. Emotion understanding allows computers to engage, interact, educate and serve us better. Since people communicate predominantly via speech, emotion classification of users' speech is the first step towards automatic emotion understanding. One of the challenges of emotion classification is that the intensity of natural emotion is not constant in time. Understanding the dynamic changes of emotion in speech is thus an important research area of emotion classification.



https://www.aldebaran.com/en/a-robots/who-is-pepper

Figure 1: An example of interaction during which emotion is a key to successful human-agent relationship

Local Dynamics of Emotion

This work focuses on local dynamics – dynamics within variable length segments of speech utterances. Local dynamics are shown to be localized[1]. They do not depend on the utterance segmentation, and they are emotionally salient[2].

This work identifies "emotion units", which are exact patterns of the emotion dynamics in speech. These patterns capture the changes of the intensity of basic emotions in time, and can be used to classify the emotions of speech utterances.

Approximating Emotion Flow

In speech recognition works, the acoustics data can be represented either by the magnitude of voice over time, or by a spectrogram, in which frequencies are the basic units of representations.

To draw an analogy, we represent emotions by a "spectrogram" in time. This representation is called Emotograms Emotograms capture the intensity of basic emotions (in this case happy, angry, sad and neutral) within each audio frame.

Figure 3: Temporal description of speech data

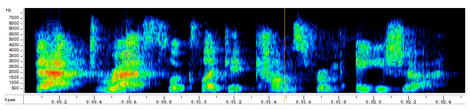


Figure 4: Frequency description of speech acoustic (spectrogram)

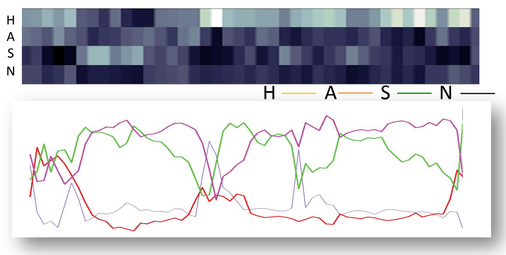


Figure 5: Emotograms: intensity of basic emotions in each speech segment (in "spectrogram" look or in magnitude look)

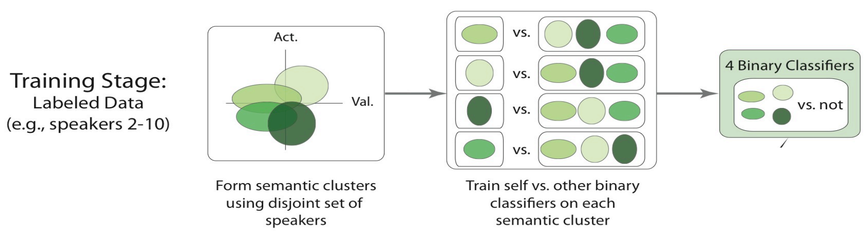


Figure 6. Training Emotograms

Emotograms are created by using a sliding window to break the training utterances into overlapping segments. Each segment is labeled with the utterance's emotion label. We then train four binary classifiers with the segments: happy vs non-happy, angry vs. non-angry, etc. When a test utterance comes in, we apply the same sliding window to the utterance, and estimate the emotion intensity in each segment. When the emotion intensity is plot against time, we generate the Emotograms.

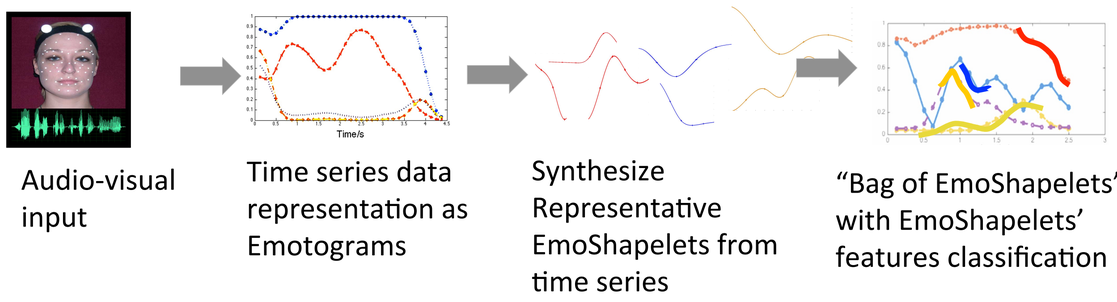


Figure 7 : Flow chart of the whole project.

References

[1]. E. Mower Provost. "Identifying Salient Sub-Utterance Emotion Dynamics Using Flexible Units and Estimates of Affective Flow." *International Conference on Acoustics, Speech and Signal Processing (ICASSP)* . Vancouver, British Columbia, Canada. May, 2013.
[2]. M. Grimm et al. "Primitives-based evaluation and estimation of emotions in speech." *Speech Communication* 49.10 (2007): 787-800.
[3]. J. Hills, J. Lines, E. Baranauskas, J. Mapp, and A. Bagnall, "Classification of time series by shapelet transformation," *Data Mining and Knowledge Discovery*, vol. 28, no. 4, pp. 851–881, 2014.
[4]. Y. Kim and E. Mower Provost. "Emotion Classification via Utterance-Level Dynamics: A Pattern-Based Approach to Characterizing Affective Expressions." *International Conference on Acoustics, Speech and Signal Processing (ICASSP)* . Vancouver, British Columbia, Canada. May, 2013.
[5]. F. Petitjean, et al., "Dynamic time warping averaging of time series allows faster and more accurate classification," *in IEEE International Conference on*, 2014.

Method

After audio-visual speech input data is represented as Emotograms, as seen in **Figure 7**. We can adopt time series data mining methods to the capture local emotion dynamics. These dynamics are called ***EmoShapelets***. Finally we use "bag of EmoShapelets" classification methods to classify test utterances.

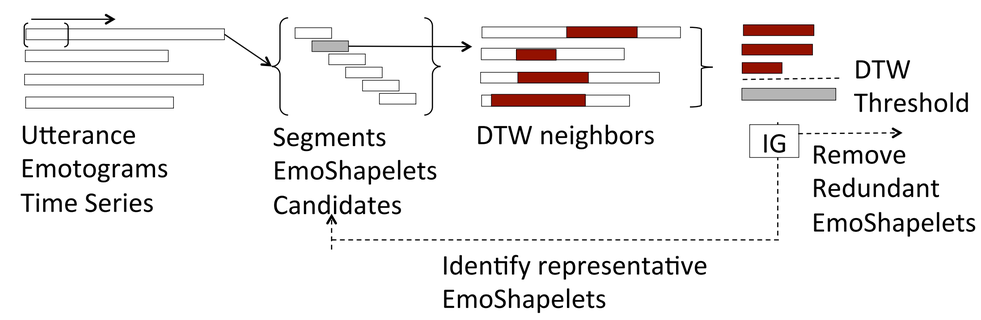


Figure 8 : Synthesizing Representative EmoShapelets from time series

EmoShapelets generation is inspired by a time series classification method called "Classification Shapelets"[3]. In the first step, we use a sliding window with length l, and 0.5 l over lap to slide through the Emotogram, and generate segments of variable lengths. These segments are EmoShaepelet candidates. The second step involves identifying "neighbors" of an EmoShapelet candidate. These neighbors are defined by segments from Emotograms that falls within certain distance threshold of the candidate. To measure distances, we adopt the technique of Dynamic Time Warping (or DTW). The third step is to use neighbors that fall within certain DTW distance threshold to update the original segment using DTW Barycenter averaging (DBA)[4]. The quality of the EmoShapelet is measured by the multiclass information gain *(maxIG in Equation 2)* of the threshold with respect to the emotion label of the neighboring segments. The threshold of the neighboring segment distances is determined by the threshold that results in maximum information gain. Non-redundant EmoShapelets generated from this process is collected and used to classify test utterances.

$$IG(e_k|newS) = H(e_k) - \sum_{s\in[newS,\sim newS]} P(S=s)H(e_k|S=s) \quad \textbf{Equation 1}$$

$$maxIG = max_{H,A,N,S} IG(e_k|newS) \quad \textbf{Equation 2}$$

Results on IEMOCAP Dataset

Classifier	EmoShapelet Property	DBA update, maxIG	no DBA update, maxIG
SVM (linear kernel)	$d_{D,M,IG}$	64.2±4.4	57.7±4.7**
k-NN (cosine distance)	$d_{D,M,IG}$	64.9±4.3	33.2±11.7**
Decision Tree	$d_{D,M,IG}$	41.2±9.2	24.3±3.3**
Naïve Bayes	$d_{D,M,IG}$	64.4±4.9	57.3±4.9**

Table 1. Classifying utterances with "bag of EmoShapelets' features", including magnitude of EmoShapelets, IG of EmoShapelets and DTW distances from EmoShapelets to utterances.

Prev. Work	Baseline Acc.
Kim and Mower Provost, 2013	63.95%
Mower Provost and Narayanan, 2011	64.20%
Mower Provost 2013	63.80%

Table 2. Baseline for dynamic methods on IEMOCAP dataset.

Scheme/EmoShapelet lengths	DBA update	1.25s	2.5s	3.75s	5.00s	0.625-5.0s	1.25-3.75s
Sum the maxIG of EmoShapelets	yes	51.9	56.1	53.6	54.8	59.7	58.7
	no	49.2	57.2	53.9	54.6	61.5	59.6

Table 3. Classifying utterances by simply summing the IG of EmoShapelets within each utterance. Results shown classification rates with different EmoShapelets lengths. This table demonstrates that EmoShapelets of different lengths are complementary.

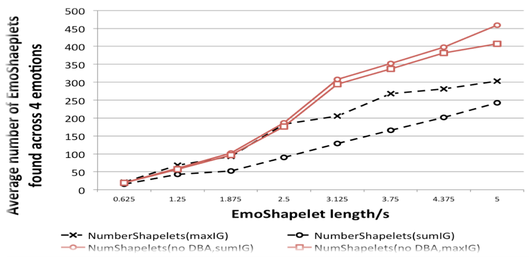


Figure 9. Number of EmoShapelets generated without DBA updates (red) vs. with DBA updates. Results demonstrated that no DBA update will result in larger number of less emotionally representative EmoShapelets collected.

Conclusions

In this work, we extract localized representative patterns from a dynamic representation of speech emotions, and name these patterns EmoShapelets. We show methods to extract variable lengths of EmoShapelets, and update the EmoShapelets using DBA. EmoShapelets can be seen as "basic units" of speech emotion and can be used to classify speech emotions to achieve comparable state-of-the-art results on IEMOCAP dataset.

Acknowledgments

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